Contributing to OAI

Robert Schmidt

March 31, 2026

CI/CD: Review and Workflow

Human Part

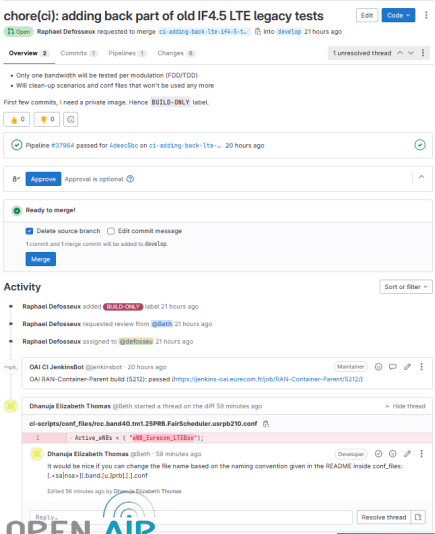
- Contributor has to sign CLA
- Senior OAI developer “volunteers” for review
- Every week: 30min slot in dev meeting to discuss merge status
- Public access to any build/test logs
- Documentation: `doc/TESTBenches.md`

Automated Part

- Apply label in MR (Doc, Build, 4G, 5G)
- Jenkins pipeline is triggered
  - Container image build
  - Static analysis
  - Testing (next slide)
- No automatic acceptance of MRs: integration branch
- Images on docker hub published by `oaisoftwarealliance` org

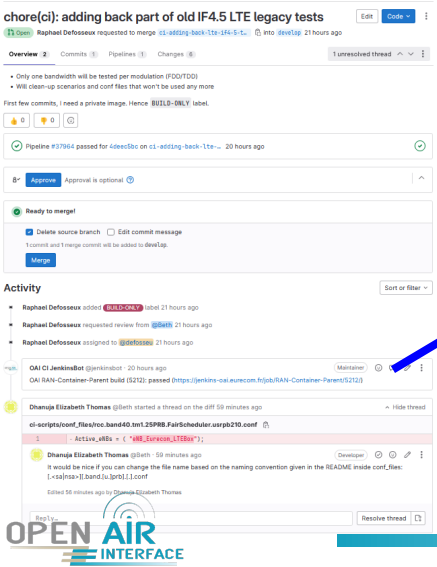
copyright @ 2026 OpenAirInterface Software Alliance

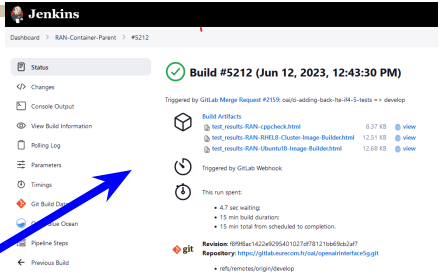
CI/CD: Review and Workflow



copyright @ 2026 OpenAirInterface Software Alliance

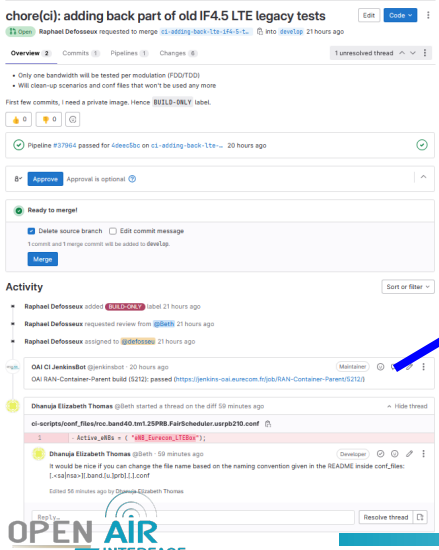
CI/CD: Review and Workflow

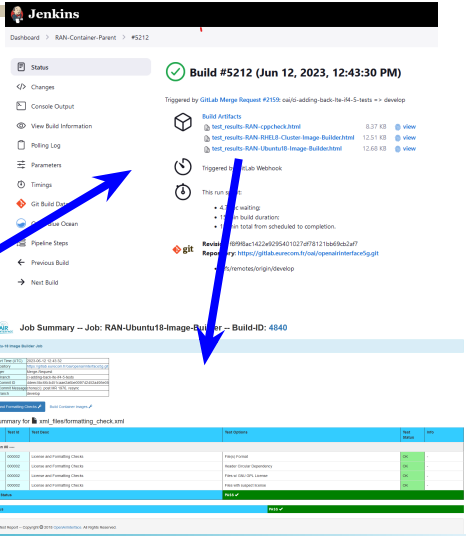




copyright @ 2026 OpenAirInterface Software Alliance

CI/CD: Review and Workflow





copyright @ 2026 OpenAirInterface Software Alliance

CI: Current pipelines

Build & Simulators

- Build: x86/ARM (v8/v9)
- Unit tests
- RFsim (scenarios…; incl. sanitizers)
- vrtsim
- Physical simulators (incl. sanitizers); AAL

Hardware-based Tests

- USRP-based, COTS UE (4G/5G)
- O-RAN 7.2 FH with VVDN/Metanoia/Benetel/LITEON
- AW2S-based with up to 16 UEs (5G)
- OAI-UE with OAI gNB OTA (4G, 5G)
- Nvidia CUDA-Accel. RAN with Quectel/OAI UE

copyright @ 2026 OpenAirInterface Software Alliance

Types of Contributions

What is a good contribution?

- Bug reports on Gitlab
- Improve existing/add new documentation
- Bug fix
- Help with review (e.g., testing)
- (New Feature)

copyright @ 2026 OpenAirInterface Software Alliance

Types of Contributions

What is a good contribution?

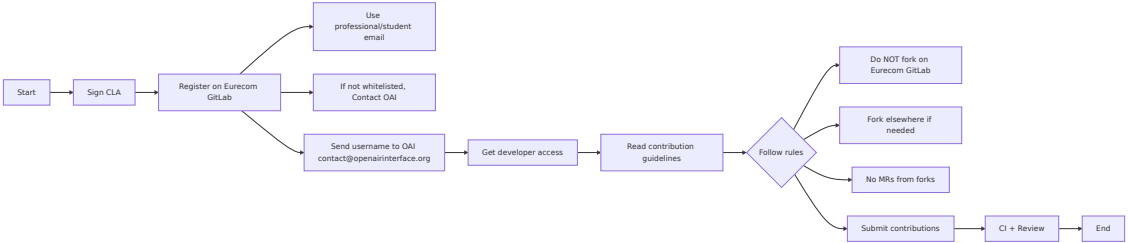
- Bug reports on Gitlab
- Improve existing/add new documentation
- Bug fix
- Help with review (e.g., testing)
- (New Feature)

What is a bad contribution?

- *Code you don't understand*
- *Too many changes*
- ?

copyright @ 2026 OpenAirInterface Software Alliance

How to contribute?



copyright @ 2026 OpenAirInterface Software Alliance

CLAs

Individual Contributor License Agreement (ICLA)

Corporate Contributor License Agreement (CCLA)

copyright @ 2026 OpenAirInterface Software Alliance

Thank you!

Q? A!

Special thanks

https://www.openairinterface.org/

copyright @ 2026 OpenAirInterface Software Alliance